MODULE 02

Data Extraction

DATA SCIENCE · LEARNING NOTES

Mohan Babu University · Course-aligned study guide

Concepts • examples • practice • revision

01. Feature selection

Learning focus. This section introduces feature selection as part of data extraction. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to feature selection. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

02. User retention analytics

Learning focus. This section introduces user retention analytics as part of data extraction. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to user retention analytics. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

03. Filter methods

Learning focus. This section introduces filter methods as part of data extraction. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to filter methods. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

04. Wrapper methods

Learning focus. This section introduces wrapper methods as part of data extraction. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to wrapper methods. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

05. Entropy and information gain

Learning focus. This section introduces entropy and information gain as part of data extraction. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to entropy and information gain. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

06. Decision tree algorithm

Learning focus. This section introduces decision tree algorithm as part of data extraction. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to decision tree algorithm. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

07. Random forests

Learning focus. This section introduces random forests as part of data extraction. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to random forests. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

08. Dimensionality problem

Learning focus. This section introduces dimensionality problem as part of data extraction. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to dimensionality problem. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

09. Single value decomposition

Learning focus. This section introduces single value decomposition as part of data extraction. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to single value decomposition. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

10. Principal component analysis

Learning focus. This section introduces principal component analysis as part of data extraction. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to principal component analysis. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

11. Feature scaling

Learning focus. This section introduces feature scaling as part of data extraction. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to feature scaling. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

12. Model interpretation

Learning focus. This section introduces model interpretation as part of data extraction. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to model interpretation. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

13. Worked example

Learning focus. This section introduces worked example as part of data extraction. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to worked example. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

14. Module review

Learning focus. This section introduces module review as part of data extraction. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to module review. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?